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Key Points:

- WRSA evolved from a prescriptive science to science of water resources systems
- WRSA has long focused on the water-related variables that are important to society
- WRSA should adopt a scientific initiative based on the prediction of water resources variables

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The future of water resources systems analysis: Toward a scientific framework for sustainable water management

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Abstract This paper presents a short history of water resources systems analysis from its beginnings in the Harvard Water Program, through its continuing evolution toward a general field of water resources systems science. Current systems analysis practice is widespread and addresses the most challenging water issues of our times, including water scarcity and drought, climate change, providing water for food and energy production, decision making amid competing objectives, and bringing economic incentives to bear on water use. The emergence of public recognition and concern for the state of water resources provides an opportune moment for the field to reorient to meet the complex, interdependent, interdisciplinary, and global nature of today's water challenges. At present, water resources systems analysis is limited by low scientific and academic visibility relative to its influence in practice and bridled by localized findings that are difficult to generalize. The evident success of water resource systems analysis in practice (which is set out in this paper) needs in future to be strengthened by substantiating the field as the science of water resources that seeks to predict the water resources variables and outcomes that are important to governments, industries, and the public the world over. Doing so promotes the scientific credibility of the field, provides understanding of the state of water resources and furnishes the basis for predicting the impacts of our water choices.

1. Introduction

Societies around the world have long struggled with the planning and management of water resources amid growing populations and resource use, competition among users, and more recently, widespread ecosystem degradation and climate change. Water resources decisions (and accidents) can spawn impacts of geologic scale including the drying of the Aral Sea [Micklin, 1988] and Lake Chad [Coe and Foley, 2001] the creation of the Salton Sea in California, the degradation of major river deltas and wetlands (e.g., Colorado, Yellow, Tigris, and Euphrates, Everglades), and alteration of streamflow through the damming of rivers [Nilsson et al., 2005]. The recognition of the complex, sociophysical nature of rising water challenges and past mistakes has led to calls for a more systematic approach to global water issues, often in the style of the Harvard Water Program (HWP) [Milly et al., 2008; Lettenmaier, 2008; Reuss, 2003].

The HWP, led by Arthur Maass, a political scientist working with engineers and economists, launched an interdisciplinary, quantitative, applied field of inquiry known as water resources systems analysis (WRSA). The field has evolved from first focusing on optimal planning of multiobjective water infrastructure investments [e.g., Maass et al., 1962; Cohon and Marks, 1975], to collaborative analysis of water management decisions [Loucks, 1995; Werick et al., 1996], to seeking better understanding of coupled human and hydrologic processes and their interactions that comprise water resources systems [Weiskel et al., 2007], all for the purpose of both informing decisions and for advancing fundamental knowledge.

WRSA is the study of water resources systems using mathematical representations of the component processes and interactions of the system to improve understanding or assist in decision-making [Loucks et al.,

1981; Loucks and van Beek, 2005]. A water resources system is the whole made from connected hydrologic, infrastructure, ecologic, and human processes that involve water. It includes biogeophysical processes, including elements of the hydrologic cycle and ecosystem functioning, and human processes, including the construction, operation, and removal of infrastructure, and other human decisions and actions such as consuming, enjoying, being harmed by, or paying for water.

A distinguishing aspect of WRSA is that components are not limited to geophysical processes, so mathematical representations of all processes, including nonphysical, are required. The interdisciplinary modeling inherent to the field represents a legacy of scholarship addressing questions recently raised in hydrology that relate to the role of human actions in the hydrologic cycle [Vogel, 2011; Sivapalan et al., 2012]. WRSA is used to test our understanding of water resources systems, assess possible changes to systems or policies and evaluate likely results of alternative decisions. WRSA focuses on water as a resource, for example for domestic use [Rosenberg et al., 2007], as an agricultural and industrial input [Rosegrant et al., 2002], and a sustainer of ecosystems [Gao et al., 2009], and also as a hazard in the form of floods and drought. Ultimately, WRSA is used to improve water planning and management decisions and is best when closely coupled with practice [Loucks, 1995].

The long history of creating models that integrate and sometimes couple physical and human systems places WRSA as a key discipline for understanding and resolving urgent water-related challenges. The field of WRSA originated to prescribe recommended solutions to large multiobjective water investment questions [Maass et al., 1962]. The HWP introduced environmental objectives and social objectives for regional development and equitable benefit distribution that better reflected societal interests than a single economic objective [Reuss, 2003]. Thus the field originated to address major water resource challenges we now recognize as sustainable water management.

Despite these promising attributes, at present WRSA suffers from low visibility among academics, lack of research funding by federal science agencies, and criticism that research findings are context specific without providing fundamental insight. A contributing factor is the focus on location-specific models and scales of analysis. This limits the generalizable insights that can be derived from many analyses. It also is an obstacle to addressing the global nature of many modern water challenges, including effects of agricultural trade and the network risk of hydrologic extremes. WRSA is widely practiced in academia and applied by operational and planning agencies and this body of knowledge is invaluable for solving the water challenges of the 21st century. Much more is needed from the field in terms of diagnosing global water challenges and recommending solutions.

In this paper, we argue that the WRSA field would benefit from better articulation of the value that these research methods provide to the scientific community in addition to the world of practice. We recognize two primary contributions: (1) providing the means by which progress in geophysical sciences (and other innovations) advance improvement in water management decisions and thus have direct value to society, and (2) providing improved fundamental understanding of water resources systems themselves which is also needed for improving water management decisions and achieving sustainable water management.

The essence of WRSA is predicting the hydrologic, socioeconomic, and environmental consequences of water management. These are the variables, such as future water availability, the economic value of water, and the reliability of environmental flows, that are important to governments, industries, and the public. The field is positioned to lead the research efforts that will advance water management to meet the challenges of the 21st century but requires a sound scientific basis to do so. In particular, the development of a scientific framework based on prediction everywhere of observable water resources variables and outcomes both hydrologic and socioeconomic, is needed to spark innovation and achieve improved understanding of water resources systems. In turn, the resultant insights will provide the basis for predicting the effects of policy choices and increase the credibility of the water management guidance offered by the field.

The purpose of this paper is to describe the field of water resources system analysis and to discuss its future. A short history of the field is presented, followed by review of a representative set of application topics. Next the need for a scientific framing for WRSA is introduced with initial thoughts on its design. The authors hope this paper engenders a rich dialog on the continued evolution and invigoration of the field.

2. Origins and Evolution of WRSA

WRSA originated for systematic evaluation of water infrastructure plans and designs. The first water planning may have served as a basis of civilization [Wittfogel, 1956; Worster, 1985] was documented by the Romans; early modeling approaches were described in the early 1800s [Lund *et al.*, 2006]. The modern field originated with the Harvard Water Program, where the analysis methods commonly applied for designing single structures were formalized and expanded. The planning of the Tennessee Valley Authority, which used a basin wide approach that included the benefits of coordinated development and operation of multiple projects, was a major influence [Borges, 1979].

The HWP introduced optimization and pioneered simulation use with computers for systematic evaluation of alternative designs for investments in water infrastructure (cited chapters). The optimization problem consisted of an objective function that specifies the goal of the planner in mathematical terms, decision variables which represent alternative choices (e.g., dam locations and sizes), and constraints, which enforced physical laws like mass balance and benefits and costs of different actions.

Although the HWP is largely remembered for applying optimization to water planning, its more general and important contribution was the concept of multiobjective analysis. Multiobjective analysis expanded the typical single economic objective of water resources planning (maximize net economic benefits) [Dupuit, 1844; Ekelund and Hebert, 1999 cited in Lund *et al.*, 2006] by adding environmental and social objectives. The argument of Arthur Maass and colleagues was that federal water investments should serve the national interests and the national interests were broader than simply maximizing net economic benefit. For example, a national interest in aiding the economic development of poorer parts of the country could be incorporated into the analysis. In attempting to address this challenge, innovation was required [Reuss, 2003]:

Questions emerged that crossed disciplinary boundaries. Were engineering solutions economically efficient and socially beneficial? How should regions be defined, i.e., culturally, economically, common natural resources, drainage basin, etc.? What should be the objectives of regional planning, and did these objectives threaten traditional institutions and life-styles?

Addressing these questions required a new way to conceptualize planning for river basin development. New kinds of models were needed that included functions that explicitly quantified these new objectives. Social objectives such as income distribution, regional development, and environmental objectives were key additions. The relative weighting of these objectives was sought from policy makers [Maass *et al.*, 1962]. These objectives remain the foundation of sustainable development and sustainable water management [c.f., *Global Water Partnership*, 2003].

WRSA ultimately was widely adopted by the U.S. Army Corps of Engineers and other federal agencies for planning new projects. In addition, the results of the program led directly to the founding of *Water Resources Research* in 1965 with first editors, Walter Langbein, a hydrologist, and Allen Knesse, a resource economist. The journal was envisioned as “the forum for furthering the spirit of the (Harvard Water) program in merging economic and physical principles for achieving efficient and effective management of the nation’s water resources” [Matalas, 2003]. Twenty percent of the first volume of WRR were WRSA papers. Since that time, the percentage of policy-oriented papers in WRR has steadily declined (Figure 1) and the scope of topics has narrowed. Where once articles related to systems analysis of water quality and wastewater problems were common, they have become rare.

Political opposition to water resources planning lessened the need for WRSA in the late 1970s [Reuss, 1992, 2003]. These developments were soon followed by the launch of “hydrologic science” [National Research Council, 1991] and diminishing federal involvement in water management, which both contributed to the end of most federal funding for research in WRSA [National Research Council, 2004; Lettenmaier, 2008].

Nonetheless, the clear distinction of WRSA from hydrologic science also created opportunity. The study of hydrology focused on understanding strictly geophysical processes and, consequently, places where the impacts of human activities were not significant, e.g., headwater catchments or modeled “naturalized flows.” This created a gap in the study of aspects of the hydrologic cycle that were affected by human

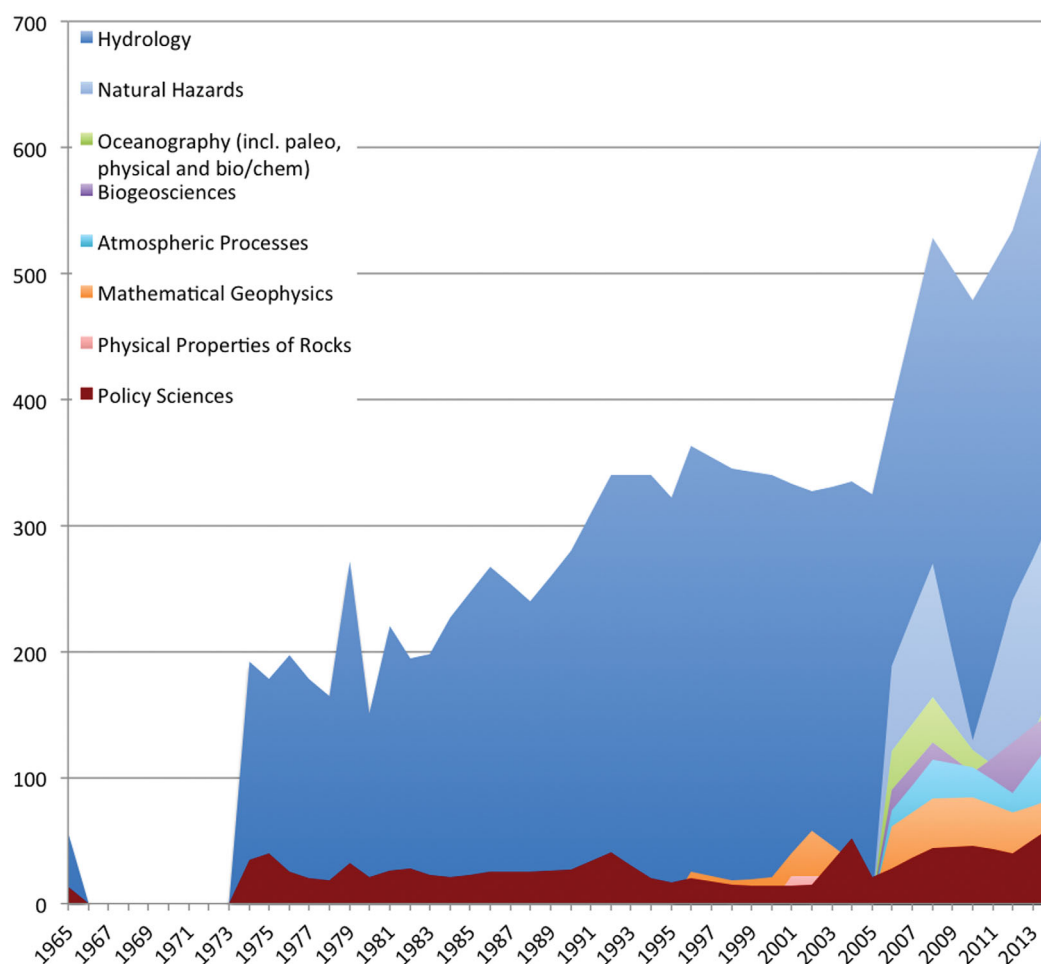


Figure 1. The number of “Policy science” (AGU Index Term) papers in WRR and the other index terms that exceed it by year. Until 1996, “Policy Science” was always the second most popular index term after “Hydrology.”

activities, which are influential or dominant in many water bodies of interest to society [Vörösmarty *et al.*, 2010]. The result was a gap between scientific advances in hydrologic science and their significance for water management.

WRSa occupies this gap between advances in hydrologic science and improved water management. In need of funding, WRSa researchers increasingly responded to the practical water management and policy problems of local and state agencies, the Bureau of Reclamation, USACE, and stakeholder groups. Planning approaches were applied and developed internationally where the need for infrastructure (and planning) was greater [e.g., Guariso and Whittington, 1987; Rogers *et al.*, 1993]. Also, the evaluation of operations and existing infrastructure took prominence over planning new development [e.g., Wurbs, 1993; Kelman *et al.*, 1990]. Finally, system analysis techniques were embedded in operational software and widely applied in operations (e.g., RiverWare [Zagona *et al.*, 2001]; RTC Tools [Schwanenberg *et al.*, 2015]; WEAP [Yates *et al.*, 2005]; and ResSim [Klipsch and Hurst, 2003]).

In addition, recognized limitations of WRSa in practice suggested a more pragmatic approach. Rogers and Fiering [1986] summarized the state of affairs in terms of two emblematic issues: (1) a lack of trust in optimal solutions due to uncertainty in various aspects of the model and (2) the recognition that no single optimum exists—that often many solutions have similar objective function values but very different implications for system design and operation. The authors suggested that optimizing models should be “softened” to allow more interactive modeling with stakeholder participation and more assessment of implications of uncertainty and sensitivity analysis of performance rather than search for optimal solutions.

The resulting evolution of methods and continued engagement with practice transformed WRSA. Three primary research themes emerged: (1) continued application to water policy and planning problems but focusing on stakeholder engagement, collaborative model development, characterization of the consequences of alternative options rather than specifying preferred options [e.g., *Palmer et al.*, 1999], and better treatment of uncertainty and incorporation of robustness concepts [e.g., *Watkins and McKinney*, 1997; *Ray et al.*, 2013], (2) application of prescriptive modeling (e.g., optimization) where well suited to support decisions that were less subjective in nature (e.g., reservoir operations, groundwater pumping, levee design, financial mechanisms) [*Mays and Tung*, 1981; *Gorelick et al.*, 1983; *Yeh*, 1992; *McKinney and Lin*, 1996; *Savic and Walters*, 1997], and (3) the use of integrated models to create mathematical representations of coupled human and hydrologic systems, such as river basin models, groundwater coupled systems, and even global agricultural production and trade models, to make projections of the future evolution of these systems [*Rosegrant et al.*, 2002; *Molle*, 2006; *Schlüter and Pahl-Wostl*, 2007; *Wainwright*, 2008; *Kanta and Zechman*, 2013].

Despite continued progress in these areas, reflection on the historical evolution of WRSA raises some concerns, including the narrowing of the field, less prominence within academia and *Water Resources Research*, and lack of federal funding for fundamental research. These concerns may be at least partially addressed by buffering the body of knowledge achieved with a stronger scientific footing for future research based on prediction as described in the remainder of this paper.

3. Current State of the Science

WRSA covers a broad range of research themes. This review covers research areas that have emerged from the origins of WRSA including the broad topic of water policy and planning, followed by examples of prescriptive modeling applications in multiobjective optimization, water infrastructure operations, and water distribution networks. These themes are united in using mathematical modeling to address societal implications of human decisions related to water management and constrained by physical laws and human institutions. This section illustrates the body of knowledge achieved in the field to date. The topic of prediction is covered in section 4.

Reviews of aspects of WRSA have appeared in the literature recently, including reviews of models used [*Labadie*, 2004], hydroeconomic modeling, [*Lund et al.*, 2006; *Harou et al.*, 2009], game theory [*Madani*, 2010], and optimization approaches [*Reed et al.*, 2013; *Maier et al.*, 2014]. These topics and are not covered here. Inevitably, the following represents a sampling of the field rather than a comprehensive review.

3.1. Water Policy and Planning

Water management activities are often expensive, capital-intensive, involve thousands to millions of users with diverse interests, and require approval from many political leaders and agencies. Policy and planning decisions are thus subject to extensive discussions and procedures, and are sometimes subject to serious analysis to develop and compare alternatives constructed to inform such discussions. Water resource systems analysis has long sought to provide more rigor, method, variety, and speed in developing water management plans and policies. Most recently, WRSA has evolved to focus on assessing alternative plans in collaboration with stakeholders in a systematic fashion using optimization and simulation methods.

Policy analysis within WRSA primarily consists of the quantitative assessment of policy impacts, assessment of alternatives, development and design of policies, and trade-off analysis. Policies here generally refer to the guidelines that prescribe, constrain, or influence water management decisions. WRSA analysis traditionally does not focus on how policies come to be but rather their impacts, although recent collaboration among economists, lawyers, political scientists, and systems analysts may shed new light on policy development [*Lund et al.*, 2010; *Hanak et al.*, 2011; *Rogers and Leal*, 2010].

Simulation modeling has become routine for analyzing water policies and almost universal in developing water plans. Simulation is used to evaluate plans or policies that are formed outside the model, typically described as “what if. . .?” analysis. Simulation applications range from major water resource system models, which include economic and environmental performance indicators to the near-universal use of hydraulic simulation models to examine the performance of flood and water distribution system alternatives. The models either include hydrologic processes, including groundwater (e.g., WEAP [*Yates et al.*, 2005]) or accept

inflows as input (e.g., ResSim [Klipsch and Hurst, 2003]). The actions of humans (e.g., reservoir operations, water use) are usually prescribed in alternatives or target objectives.

While simulation has been largely victorious as an analysis approach, and is now widespread and expected [Lund, 2010], water policy and planning applications of system optimization analysis remain less common outside of academia. Optimization models are prescriptive models used to create plans or policies (quantified as decision variables in an objective function) that achieve some objective. In flood management, risk-based optimization is widely used in the Netherlands to establish flood safety levels for major levee systems [Eijgenraam *et al.*, 2014] and recently in the UK [Environment Agency, 2014]. The greatest official use of optimization for policy and planning problems has been where there is substantial political consensus on performance objectives or especially, a single objective (e.g., minimize net costs), as is more common for water distribution, hydropower, and levee systems.

Formal system optimization also raises insightful questions and suggests promising solutions for specific systems, often raising questions and solutions without prior political attention or support. For example, in California optimization models have been pivotal in providing insights that had not arisen in lengthy agency or stakeholder input to public and policy discussions [Lund *et al.*, 2010; Hanak *et al.*, 2011; Draper *et al.*, 2003].

Aside from routine hydropower applications, hydroeconomic optimization has been used occasionally, but usefully, for academic and policy studies. These have included restoration of Hetch Hetchy Valley in Yosemite National Park, a long-term cause celebre among some environmentalists, was raised in prominence by an academic optimization model [Null and Lund, 2006] which indicated that San Francisco's water supply could be sustained without a reservoir in Hetch Hetchy Valley. This modeling instigated a Pulitzer Prize winning series of editorials on restoring Hetch Hetchy, which led to a major study by the State of California and a vote of the people of San Francisco (which voted not to further study the matter).

Some models use optimization to simulate the decisions related to human actions within a simulation-based planning tool, rather than as an optimized basis for developing plans and design decisions. Many models use optimization to simulate water use and reservoir release decisions according to prescribed priorities (WEAP, CalSim, OASIS, Vista DSS, Riverware). Hydroeconomic modeling sometimes uses optimization in a similar simulation role, where human actions are estimated as decision variables that maximize economic outcomes [Rosenberg *et al.*, 2007]. Hydroeconomic optimization is also widely used for modeling water markets, market-like mechanisms, and innovative applications of financial mechanisms with water supply systems [Harou *et al.*, 2009; Vaux and Howitt, 1984].

Recognizing stakeholders' preference to have control, and in cases where stakeholders are unlikely to agree to objectives, collaborative modeling has gained traction [Langsdale *et al.*, 2013]. Collaborative modeling involves the codevelopment of a water resources systems model with stakeholders and a focus on decision support for contributing to a more complete understanding of the decision or problem being addressed [Palmer *et al.*, 2013]. They may use simulation or optimization. Collaborative modeling methods work from the hypothesis that better understanding of water challenges, achieved through an open process for model development, enhances the chances that solutions will be identified and accepted by stakeholders. Shared Vision Planning [Palmer *et al.*, 2013] is a prominent example, adopted by the USACE and applied in basins around the world.

WRSAs are widely used to assess proposed new infrastructure developments in major river basins around the world, especially in transboundary settings. Examples include studies of the Nile [Jeuland and Whittington, 2014], Indus [Yu *et al.*, 2013; Yang *et al.*, 2014], Niger [Ghile *et al.*, 2014], the Zambezi [World Bank, 2010a], Kabul [World Bank, 2010b], Ganges [southasiawaterinitiative.org], and Indus [Yu *et al.*, 2013], often building on earlier studies of the same basins.

Assessing the risks of climate change and planning for adaptation to its anticipated effects has been an important application area for WRSAs since at least the 1990s [Frederick and Major, 1997; Rogers, 1994; Stakhiv, 1998] and includes development and application of decision analysis [Hobbs *et al.*, 1997; Brown *et al.*, 2012] and creative responses to uncertain probabilities [Lempert *et al.*, 2006; Hall *et al.*, 2012; Jeuland and Whittington, 2014] in addition to local studies that are too numerous to mention. Finally, in a comprehensive display of planning applications, the U.S. Bureau of Reclamation used a wide range of WRSAs techniques

(in addition to hydrology and climate science) in the Colorado River Basin Water Supply and Demand Study [US Bureau of Reclamation, 2012].

Water planning applications of WRSA also applied to water demand modeling and management [Chicoine *et al.*, 1986; Hewitt and Hanemann, 1995; Schoengold *et al.*, 2006], groundwater management, and conjunctive use [Rogers and Smith, 1970; Gorelick, 1983; Bredehoeft and Young, 1983; Provencher and Burt, 1994; Emch and Yeh, 1998; Karamouz *et al.*, 2004; Mulligan *et al.*, 2014] and ecosystem restoration [Sale *et al.*, 1982; Cardwell *et al.*, 1996; Shiau and Wu, 2007; Vogel *et al.*, 2007; Steinschneider *et al.*, 2013].

3.2. Addressing Trade-Offs Among Multiple Objectives

At its inception by Maass *et al.* [1962], WRSA represented arguably one of the first instances of advocating for explicitly characterizing the trade-offs between conflicting economic, social, and environmental performance objectives when evaluating the costs and benefits of large-scale water resources infrastructure investments [Banzhaf, 2009]. In simple terms, multiobjective optimization seeks to identify those decisions or actions whose performance in one objective can only be improved by degrading performance in one or more remaining objectives (i.e., the Pareto optimal trade-offs) [Pareto, 1896]. It is worth noting that the initial efforts of the Harvard Water Program in the 1950s exploring multiobjective water resources systems design were very strongly linked to the emerging solution tools and theoretical understanding of multiobjective programming in that same decade [Koopmans, 1951; Kuhn and Tucker, 1951]. This close coupling between water resources systems and advances in multiobjective solution techniques has remained consistently true to present time [see Haimes and Hall, 1977; Maier *et al.*, 2014; Nicklow *et al.*, 2010; Reed *et al.*, 2013; Yeh, 1985 for the relevant discussions]. In fact, Water Resources Research has since its very earliest issues been a forum for the advances and debates in the area of multiobjective optimization [e.g., see Cohon and Marks, 1973, 1975; Cohon *et al.*, 1979; Major, 1969; Monarchi *et al.*, 1973; Vemuri, 1974 early studies].

Cohon and Marks [1975] proposed a classification of multiobjective approaches based on when they require decision makers to express their preferences for how conflicting objectives should be balanced: (1) a priori articulation of preference through classical multicriterion decision-making techniques [e.g., Chankong and Haimes, 1983], (2) a posteriori articulation of preferences after explicitly generating trade-offs [e.g., Cohon *et al.*, 1979], and (3) progressive articulation of preference during the search process [e.g., Monarchi *et al.*, 1973]. This classification of multiobjective techniques remains among the most dominant across all technical domains of application [see Coello Coello *et al.*, 2007 for detailed review]. Recent reviews of the last 20 years of the water resources systems literature [Maier *et al.*, 2014; Nicklow *et al.*, 2010] highlight that there has been a strong shift away from a priori multiobjective techniques that require explicit specification of priorities, goals, or weights so that multiobjective problems can be transformed and solved using single objective solution techniques. There has been since the 1990s a dramatic expansion of the use of a posteriori trade-off analysis across all water systems application areas [Nicklow *et al.*, 2010; Reed *et al.*, 2013]. For example, there are now water agencies that are independently incorporating advanced trade-off analyses as part of their planning processes [Groves *et al.*, 2008; U.S. Bureau of Reclamation, 2012; Basdekas, 2014]. Although there are examples of progressive articulation of preference in multiobjective optimization [Babbar-Sebens and Minsker, 2010; Singh *et al.*, 2008, 2010], these approaches are still an area of emerging research. These represent another way in which the first phase of WRSA was softened in response to earlier critiques [Rogers and Fiering, 1986].

Exponential growth rates in our computational capabilities as well as the growing recognition that many current and future water systems planning decisions require robustness to deeply uncertain projections of future states-of-the-world, serve as both opportunities and challenges to exploiting multiobjective optimization in our field. Recent computational benchmarks [Hadka and Reed, 2015; Reed and Hadka, 2015; Reed *et al.*, 2013] highlight that for the first time in our field an array of solution tools exist now that can operationally address 2–10 conflicting objectives, severe problem traits (e.g., nonlinearity, severe constraints, stochastic objectives, mixed real and integer decisions, etc.), and extreme computational demands.

These capabilities come with challenges. As noted very clearly by Brill *et al.* [1990], decision makers preferences can evolve opportunistically when they are allowed to see the broader trade-offs that emerge with additional decision relevant objectives. In a broad sense, this insight means that we must be careful to explore the space of newly possible problem formulations [Kasprzyk *et al.*, 2012]. Moreover, it will be critical to

evaluate which candidate formulations will generate alternatives that are sufficiently adaptive and robust to errors in projections of future conditions that impact water resources systems [Herman *et al.*, 2014; Kasprzyk *et al.*, 2013]. Last, current and future advances in multiobjective optimization require major advances in human-computer interaction, visualization of complex system dynamics, improved strategies for communicating uncertainty, and the operational exploitation of the alternative forms of parallel computing that are possible in current and future computing platforms.

3.3. Water Resources Operations

Water systems operations analysis is applied to time frames from a few minutes to a year; it proceeds from current initial conditions, considers unknowns such as inflows and demands, and produces an operating plan. For short timescales—a few minutes to one day—the plan is implemented immediately. For longer time frames such as an annual operating plan, critical operating decisions such as this year's water allocation, and power generation commitments are made and may have economic implications for various sectors such as agriculture, energy, navigation, and recreation, and affect how well environmental objectives can be met. The results influence the flow of water in rivers around the world.

The academic community has produced numerous systems analysis techniques and models for operations [Yeh, 1985; Simonovic, 1992; Wurbs, 1991, 1993; Labadie, 1997, 2004]. Although optimization has been extensively used in operational planning, i.e., developing operating criteria [e.g., Lund and Ferreira, 1996; ReVelle, 1999], most operations models implement these criteria via simulation driven by prioritized rules, e.g., U.S. Army Corps of Engineers ResSim [Klipsch and Hurst, 2003]. The requirement for operations to quickly produce a reliable operating solution that can be understood by the operators and stakeholders favors simulation over an optimization problem for which both formulation and solution may be time consuming, difficult to comprehend, and require skills beyond those of typical system operators. Simulation is possible in most cases because conflicting objectives have been resolved in planning phases; it is desirable because operators want an unambiguous solution.

A notable exception is hydropower-dominated systems where economic incentive justifies optimizing the use of operational flexibility for maximizing the value of hydropower subject to nonpower constraints and objectives. For example, Tennessee Valley Authority, Bonneville Power, Hydro Quebec, and numerous other utilities that manage water systems use systems analysis tools such as RiverWare [Zagona *et al.*, 2001], RTC-Tools [Schwanenberg *et al.*, 2015], and Vista DSS (Hatch Ltd.). Significant investment due to the high value of hydropower makes this the most vibrant area for continued improvement of systems analysis tools in water systems operations.

Uncertainty of future hydrologic conditions, from days to weeks to seasons and years, is the primary challenge in operations. Forecasting skill has improved in recent years due to technological and scientific advances including the National Weather Service's Community Hydrologic Prediction System (CHPS) and the U.S. Army Corps of Engineers' Corps Water Management System (CWMS). These systems integrate data and hydrologic models with simulation models to support near-term operations with risk-based approaches such as Monte Carlo and Extended Streamflow Prediction (ESP). Risk and uncertainty are much more difficult to incorporate into optimization models, traditionally done through stochastic linear or dynamic programming and recently ESP in mixed simulation and optimization models [Alemu *et al.*, 2011]. This remains an area ripe for advancement to improve the quality of operational forecasts and quantified risks and uncertainties.

The use of forecasts is an area where hydrologic science and the WRSA community must advance together. The research gap that persists between a forecast usually provided by hydroclimatic communities and the actual use of the forecast in water operations is an opportunity for WRSA [Simonovic and Burn, 1989]; however, many impediments are institutional [Rayner *et al.*, 2005] as well as related to forecast skill [Whateley *et al.*, 2014].

The need for more system analysis in water operations will be driven by more intense competition for limited supplies, necessitating efficient allocation and trade-offs among competing objectives. Also, hydrologic nonstationarity and greater variability associated with climate change will necessitate constant updates to optimized system rule parameters, e.g., optimal hedging rules [Draper *et al.*, 2003]. Improving system

analysis tools to be faster, more robust, easier to configure and more transparent will open the door to many more such applications in water operations.

3.4. Water Distribution Systems

Water distribution systems are complex assemblies of hydraulic control elements connected together to convey quantities of water from sources to consumers at constrained pressures and water quality requirements. A water distribution network behavior is governed by: (1) the physical laws which describe the flow and water quality relationships in the pipes and through the hydraulic control elements, (2) the consumer demands, and (3) the system's layout. Water distribution systems planning and management problems, such as least cost design of water networks, use WRSA to find the water distribution system's component characteristics (e.g., pipe diameters, pump heads and maximum power, reservoir storage volumes, etc.), which minimize the system capital and operational costs, while maintaining the system hydraulic laws and satisfying constraints on quantities and pressures at the consumer nodes. These problems are especially relevant to established utilities facing rising infrastructure maintenance costs and expanding seeking utilities seeking resilient and economic solutions, including decentralized approaches.

Management problems associated with water supply systems can be generally classified into: (1) layout (system connectivity/topology); (2) design (system sizing given a layout); and (3) operation (system operation given a design). A practical problem utilities face now is adapting an existing network to meet changing demands and risks in optimal fashion. In addition, problems related to aggregation, leakage, maintenance, reliability, unsteady flow, and security can be identified for gravity, and/or pumping, and/or storage branched/looped water distribution systems. Flow and head, or flow, head, and water quality can be considered for one or multiple loading conditions, taking into consideration inputs/outputs as deterministic or stochastic variables. In fact any combination of the categories can form a problem related to water supply management (e.g., design + flow, head, quality + unsteady).

Methods for solving water distribution system management problems used linear/nonlinear optimization schemes which were limited by the system size, the number of constraints, and the possible number of loading conditions. Decomposition methods [Alperovits and Shamir, 1977; Quindry *et al.*, 1979, 1981; Kessler and Shamir, 1989; Eiger *et al.*, 1994; Ostfeld and Shamir, 1996], for example, where one of the early traditional schemes for least cost design of water distribution systems. Those methodologies were limited in the number of loading conditions that can be considered and to converging into, most likely, local optima.

In recent years the capabilities of solving water distribution systems optimization problems have improved dramatically with the utilization of genetic algorithms [Holland, 1975]. Simpson *et al.* [1994] were the first to use a genetic algorithm for water distribution systems least cost design. They applied and compared a genetic algorithm solution to the network of Gessler [1985], as well as to enumeration and nonlinear optimization. Savic and Walters [1997] used genetic algorithms to solve and compare solutions of the one-loading gravity systems of the Two Loop Network [Alperovits and Shamir, 1977], the Hanoi network [Fujiwara and Khang, 1990], and the New York Tunnels system [Schaake and Lai, 1969].

Along with the employment of heuristic methods such as genetic algorithms, multiobjective approaches are common for water distribution systems management. In reality, management problems, such as the design problem of a water distribution system, involves competing objectives (e.g., minimizing cost versus maximizing reliability). Halhal *et al.* [1999] were the first to introduce a multiobjective scheme to solve a water distribution systems management problem. Minimizing network cost versus maximizing the hydraulic benefit served as the two conflicting objectives.

Research in water distribution systems is dramatically changing: from a distinct framework of data collection to a continuous transparent structure. With multiple types of sensor data at multiple scales, from embedded real-time hydraulic and water quality sensors to airborne and satellite-based remote sensing, integrating these capabilities into new tools for decision support for water distribution networks is a major challenge.

There is a new reality in which a state of overflowing information is replacing the constraints of data scarcity. New tools for data screening, algorithms, and metamodeling constructions, as well as computational efficiency are anticipated to dominate all future developments of water distribution systems analysis.

Table 1. Variables for Prediction in Water Resources Systems Analysis

Prediction in Water Resources Systems
Streamflow in basins with human activity present
Groundwater elevation and flow
Reservoir storages and releases
Growth and change in water use in various sectors
Costs and Benefits of water infrastructure investments
Economic value of water
Water-related employment
Agricultural production
Flood damages
Water quality
Ecological status of aquatic and riparian environment
Waterborne disease occurrence

4. The Future: Predicting Water Resources Variables Everywhere

For WRSA to continue to provide service to society, it must demonstrate value to the water management community and to the scientific community. Doing so requires focusing on the societal challenges that must be solved rather than the methodological peculiarities that can be solved. The future of WRSA depends on recognizing and engaging in the water challenges where understanding of a single component process is insufficient for understanding the system as a whole. WRSA meth-

ods should be applied to reveal fundamental insights about these challenges in addition to seeking solutions for specific instances.

WRSA researchers have traditionally sought validation through the uptake of their model results by decision makers [Rogers and Fiering, 1986] and seeking better impact drives innovation [Reed *et al.*, 2013]. However, the field of WRSA may be better served by also seeking to provide value to the scientific community. A starting point is seeking validation through a systematic predictive and evaluation framework that more clearly demonstrates gaps and gains in understanding and highlights improvement through innovation. Thus the future challenge for researchers is twofold: demonstrating that WRSA provides improved understanding of water resource systems and applying the improved understanding to achieve better water management.

We propose a WRSA community project of predicting *everywhere* water resources state variables and outcomes, in short, the hydrologic, socioeconomic, and environmental consequences that are important to governments, industries, and individuals. The project provides a pathway to improved understanding of water resources systems, recognition of knowledge gaps and data needs, and a mission around which to organize coordinated research efforts. The purpose of this effort are to: (1) provide evidence of the scientific validities of the models used, (2) provide scientific explanation of the status of water resources systems, and (3) provide the basis for predicting the effects of future exogenous factors and policy choices. Together these objectives serve to advance the practice of water resources management to meet the global water challenges of the 21st century.

Prediction of water resources state variables and outcomes (such as those listed in Table 1) everywhere requires global modeling efforts comparable to climate modeling and integrated assessment modeling. The primary weakness of current global water modeling efforts is the lack of dynamic representation of water management, including infrastructure operations, water use decisions, and responses to the scarcity value of water. These are critical to correctly diagnosing and prescribing solutions to problematic water conditions around the world. They are also all routinely addressed at local and regional scales in WRSA. A key challenge is deriving understanding across scales, with coarse resolution at global scales ultimately linking to more detailed modeling at the local scale that is currently the focus of WRSA.

Seeking to maximize the natural interests in water investments spawned the interdisciplinary research and innovations that launched WRSA. Similarly, a research program focused on predicting water resources variable everywhere will face important research gaps requiring new collaborations and methods. An initial list of gaps in knowledge include:

1. The behavioral economics and political science of how water is actually allocated in practice,
2. The long-term socioeconomic effects of water investments and policy choices,
3. Prediction and valuation of ecological responses to alternative water use, flow, and investment regimes,
4. Feedback effects of water use on local hydrology and hydrometeorology,
5. Predicting observed reservoir operations in locations without knowledge of operating policies, and
6. Predicting the evolution of water demand.

This is a fortuitous moment to strengthen WRSA research credibility through a predictive framework. In particular, there is wide recognition that interdisciplinary modeling approaches that include human actions are

needed to better understand and predict the movement of water through the biosphere [Lall, 2014; Thompson et al., 2013; Vogel, 2011; Sivapalan et al., 2012; Wagener et al., 2010]. WRSA research has long produced models of water and human interactions that incorporate social, political, and legal processes and has much to offer the scientific community here, including the hydrologic and climate science communities. WRSA methods typically employ simplified representations of hydrology and more explicitly represent human actions, with coupled physically based groundwater modeling generally an exception. In these simplified ways, WRSA has addressed the coupling of human-hydrologic systems at least since its inception. WRSA also provides a well-established link with practitioners built through decades of collaboration that is needed if new findings will yield improved water management.

WRSA can gain from predictive frameworks that are common place in hydrology [Blöschl et al., 2013]. Water resources planning is ultimately prediction of the performance of plans and policies [Loucks and Beek, 2005]. Yet typical WRSA predictions are *sui generis* and localized, often used as transfer functions for exogenous factors like climate change projections or seasonal climate forecasts. To answer 21st century water challenges, a global perspective is needed [Lall, 2014].

A major challenge for prediction is the inherent complexity of water resources systems, due especially to uncertainty in human behavior and political processes [Loucks, 1992]. Characterizing their behavior, providing better understanding of the system through its emergent properties and providing probable bounds on the future evolution of such systems may be the most plausible predictive objective [Folke et al., 2002]. Failures in prediction will be commonplace but useful for learning and to identify the need for and benefit from improved observations [e.g., Braden et al., 2013] and model formulations.

Closer observation of modeling performance in a predictive framework should lead to better characterization of predictive uncertainty, which is common in hydrology but rare in WRSA where uncertainty is surely underestimated. There's a notable lack of historical analysis of the performance of observed water resources plans [Whittington and Zhao, 2015]. One illustrative exception [Kline and Moretti, 2014, cited in Whittington and Zhao, 2015] reviewed the economic value of the TVA and found broad benefits to the basin wide planning approach applied there.

New emphasis on a predictive framework does not imply abandoning close integration with the water practitioner community. In fact, the understanding gained through collaborative experience is an essential strength of the field and a substantial advantage for improving prediction within water resources systems. Likewise, a predictive orientation of WRSA will strengthen the credibility and potentially the accuracy of the advice and guidance that is needed to achieve sustainable water management.

5. Conclusion: The Future of Water Resources Systems Analysis

The evolution of WRSA from a regional infrastructure planning approach to a broader study of water and society has produced: (1) a field immersed in the practical issues of water management and (2) a field uniquely equipped to describe and predict the water resources future that government, industries, and people seek. Together these attributes serve as the basis for a science of sustainable water management to meet the global water challenges of the 21st century.

Continued evolution is needed for WRSA to successfully address the pressing water challenges faced by society today. The field must continue to engage with the most difficult problems, marked by competing objectives, multidisciplinary processes, and amorphous values; in short, problems in their true, unreduced complexity. Yet they must be engaged using formal methods, such as the predictive framework described here, that facilitate the documentation of fundamental and general insights about the nature of these challenges. Given the awareness of water issues around the world, there is an opportunity for WRSA to articulate a vision for problem oriented, practice-based yet fundamental research. Adoption of a scientific framework based on prediction of water resources variables provides an organizing principle for fundamental research that will serve and benefit from engagement with practice. It will improve the scientific credibility of methods used, provide general insights into the status of water resources systems, and develop a scientific basis for forecasting future effects of water management choices.

Crafting a balance between research for basic understanding while providing solutions to contemporary water problems is a key challenge and opportunity for WRSA. Engaging with practice is the source of

inspiration and reveals the needs for innovation that promotes research. Advancements in understanding or prediction of theoretical benefits have little value to the water community and to society if they do not improve practice. A more formal approach to prediction and characterization of uncertainty should yield benefits in terms of improved accuracy of guidance for water management but may also make communication with practitioners more complicated. Nonetheless, without a boarder scientific framing, WRSA can continue to focus on local solutions but the full potential of the field to provide broad understanding of water challenges and yield solutions will not be realized.

Finally, while water challenges continue to arise and intensify, the ranks of WRSA specialists are declining in academia [Lettenmaier, 2008]. Junior academics lack opportunities to compete for prestigious fundamental science awards. NSF has not had a major research program in water resources for decades and programs such as Water Sustainability and Climate place primacy on interactions between traditional disciplines rather than interdisciplinary fields like WRSA. A stronger demonstration of the value provided by WRSA to the scientific community may provide a stronger argument in support for basic research funding. Regardless, federal funding for research into water issues is unlikely to be fruitful if it leaves out the primary research field that addresses these issues.

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